Statistical Learning and Data Analysis Module B

Linear Regression Models

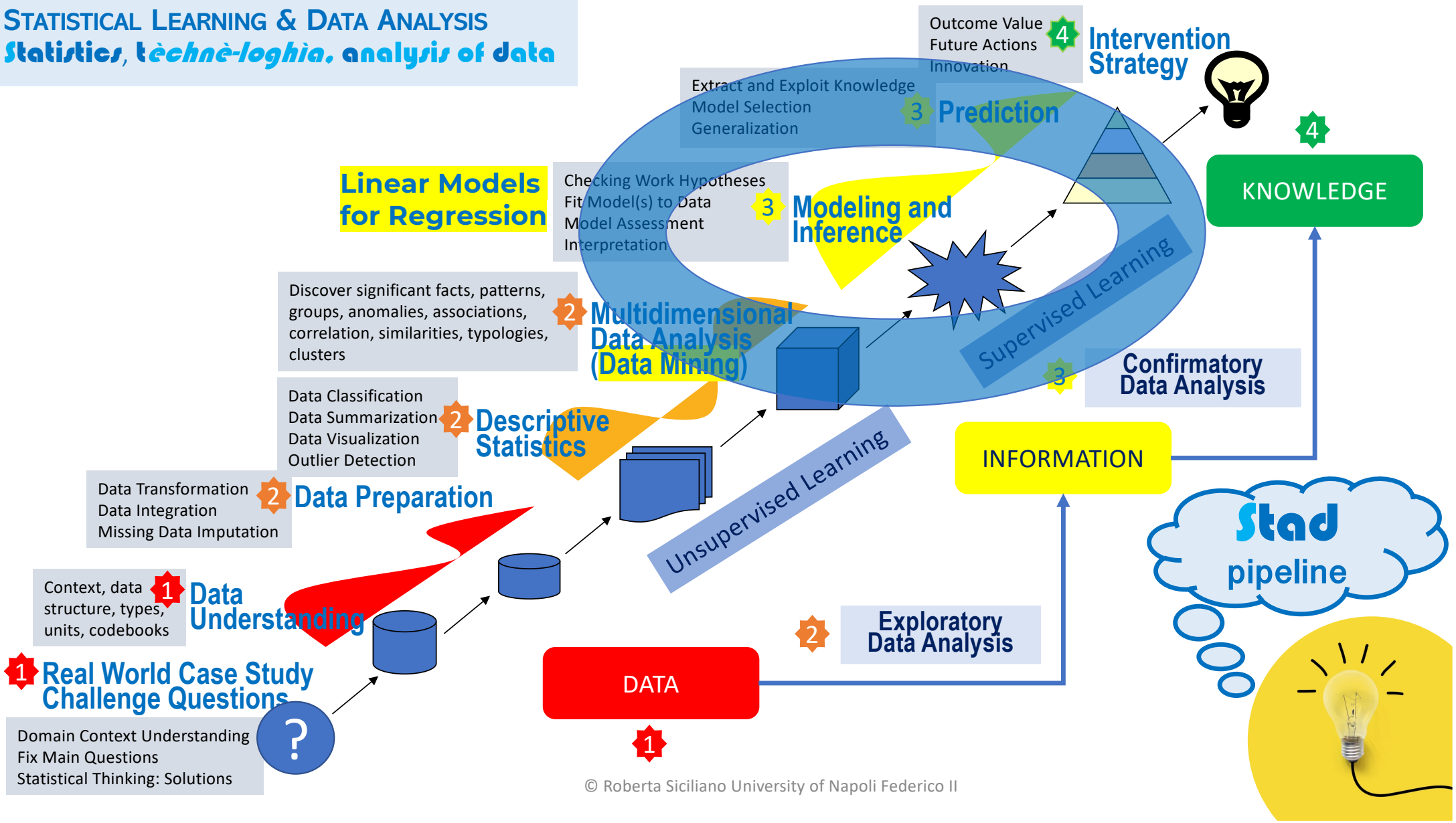
Roberta Siciliano

Statistics, tèchnè-loghìa, analysis of data

DIETI, University of Napoli Federico II

Statistical Learning & Data Analysis

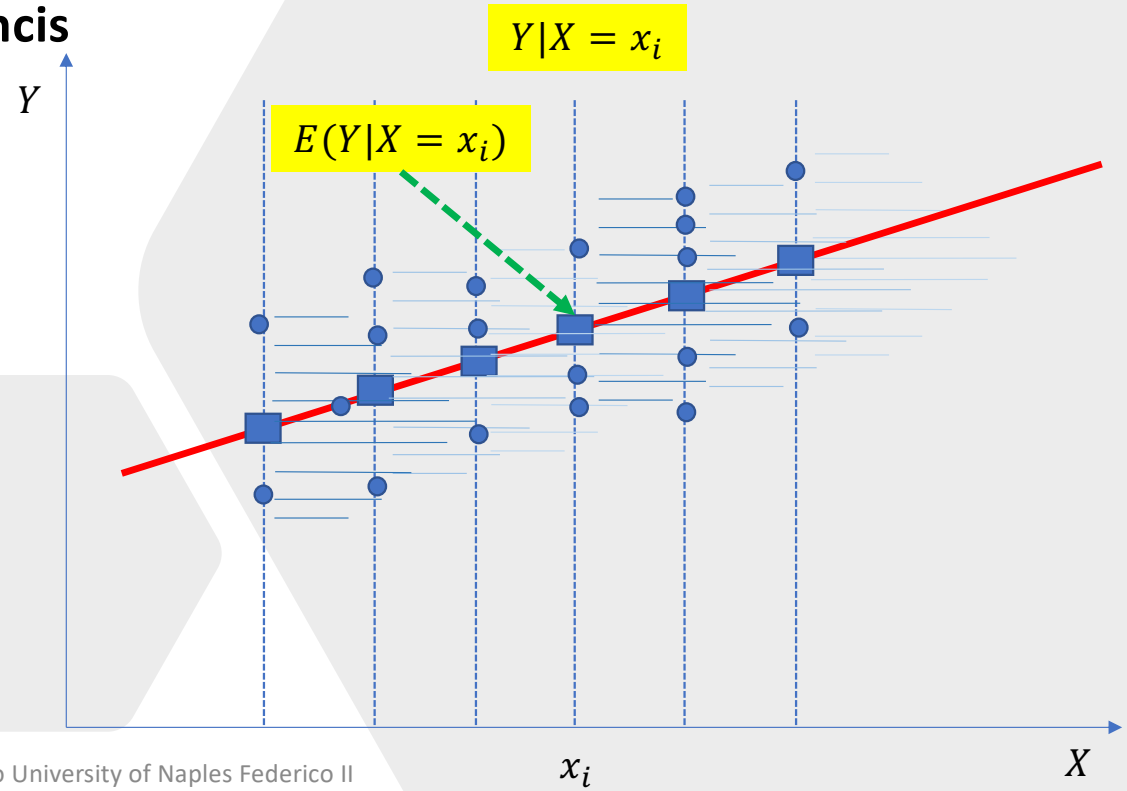
Statistics, tèchnè-loghìa, analysis of data



© Roberta Siciliano University of Napoli Federico II

Regression towards mediocrity in hereditary stature (1886) – **Sir Francis Galton**

A thought



© Roberta Siciliano University of Naples Federico II

Contents

- Statistical Model Identity
- Linear Regression
  - Assumptions
  - Estimation
  - Testing procedures
- Prediction
- Regression Diagnostics

© Roberta Siciliano University of Naples Federico II

Statistical Model Identity

- Let Y be the response variable to be explained by one predictor X.
- The Statistical Model Identity is:

$$y = f(x) + \varepsilon$$

- $f(x)$ is the fixed but unknown function providing the systematic information that X provides about Y;
- ε is the "Error" component which includes what cannot be controlled by the model.

➢ **There is a "true" model generating the data and often is a truly "best" way to analyze the data.**

© Roberta Siciliano University of Naples Federico II

Regression Analysis

- Regression means that the model systematic component is the expectation of the conditional variable $Y|X = x$.
- The statistical model can be specified as:

$$y = E(Y|X = x) + \varepsilon$$

✓ Y is the response variable of **numerical type**

✓ X is the set of predictors of *any type*

✓ ε is a random variable which satisfies "classical" hypotheses.

- The model is *linear* when $E(Y|X)$ is *linear* in the parameters.

© Roberta Siciliano University of Naples Federico II

Choice of the linear model for regression

Experimental Data

- ANOVA (Analysis of Variance) Model
  - ✓ **One-Way ANOVA: One categorical predictor, called "factor"**
  - ✓ **Two-Way ANOVA: Two factors**
- CANOVA (Covariance Analysis) Model
  - Mixed predictors
- MANOVA Model
  - Multivariate ANOVA Model for more numerical response variables

Observational Data

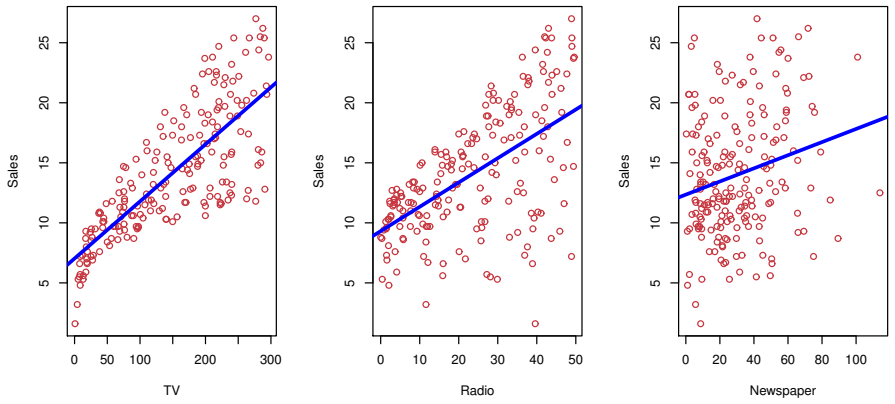
- Linear Regression Model
  - ✓ **Simple linear regression: One numerical predictor**
  - ✓ **Multiple linear regression: More numerical predictors**
  - ✓ **Linear regression with dummy variables (0/1):**
    - *Only categorical predictors recoded as dummy variables (as an alternative to ANOVA model)*
    - *Mixed predictors*
- Multivariate linear regression
  - More numerical response variables

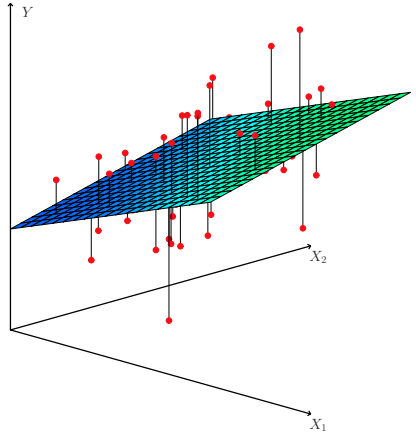
© Roberta Siciliano University of Naples Federico II

Linear Regression Model: *simple vs multiple*

- **Advertising data set**: Sales of a product (*expressed in (thousands of units*) in 200 different markets, along with advertising budgets (*expressed in thousands of dollars*) in each of those markets for three different media: TV, radio, and newspaper.

- How to understand variation of Sales (to predict Sales) of the product jointly considering the three media budgets?





© Roberta Siciliano University of Naples Federico II

Fewer important questions

- Is there a relationship between advertising budget and sales?
- Given a certain advertising budget, can we predict sales with high level of accuracy?
- Which media contribute to sales?
- How accurately can we estimate the effect of each medium on sales?
- How accurately can we predict future sales?
- Is the relationship linear?
- Is there synergy among the advertising media?

© Roberta Siciliano University of Naples Federico II

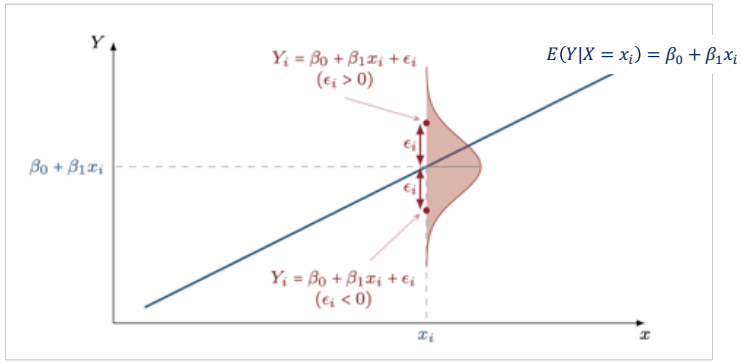
Simple Linear Regression

- **Simple Regression model:** the numerical response variable Y depends on average from one numerical predictor X:

$$y_i = E(Y|X = x_i) + \varepsilon_i \qquad i = 1, \dots, n$$

- **Linear:** the model is correctly specified by a linear function

$$E(Y|X = x_i) = \beta_0 + \beta_1 x_i$$



- **Error** satisfies classical hypotheses:

✓ *Zero mean:* $E(\varepsilon_i) = 0 \Rightarrow E(y_i) = E(Y|X = x_i) = \beta_0 + \beta_1 x_i$

✓ *Homoscedasticy:* $var(\varepsilon_i) = \sigma^2 \Rightarrow var(y_i) = \sigma^2$

✓ *Uncorrelated:* $cov(\varepsilon_i, \varepsilon_j) = 0, i \neq j$

✓ *Uncorrelated with predictors:* $cov(\varepsilon_i, x_i) = 0$ or *the predictor X is deterministic* with $var(X) \prec \infty$

- **For testing**, additional hypothesis: $\varepsilon_i \approx N(0, \sigma^2)$

© Roberta Siciliano University of Naples Federico II

Ordinary Least-Squares estimation

Minimizing the sum of the squared errors

$$\min_{\beta_0,\beta_1} Q(\beta_0,\beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

Ordinary Least-Squares (OLS) estimates:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

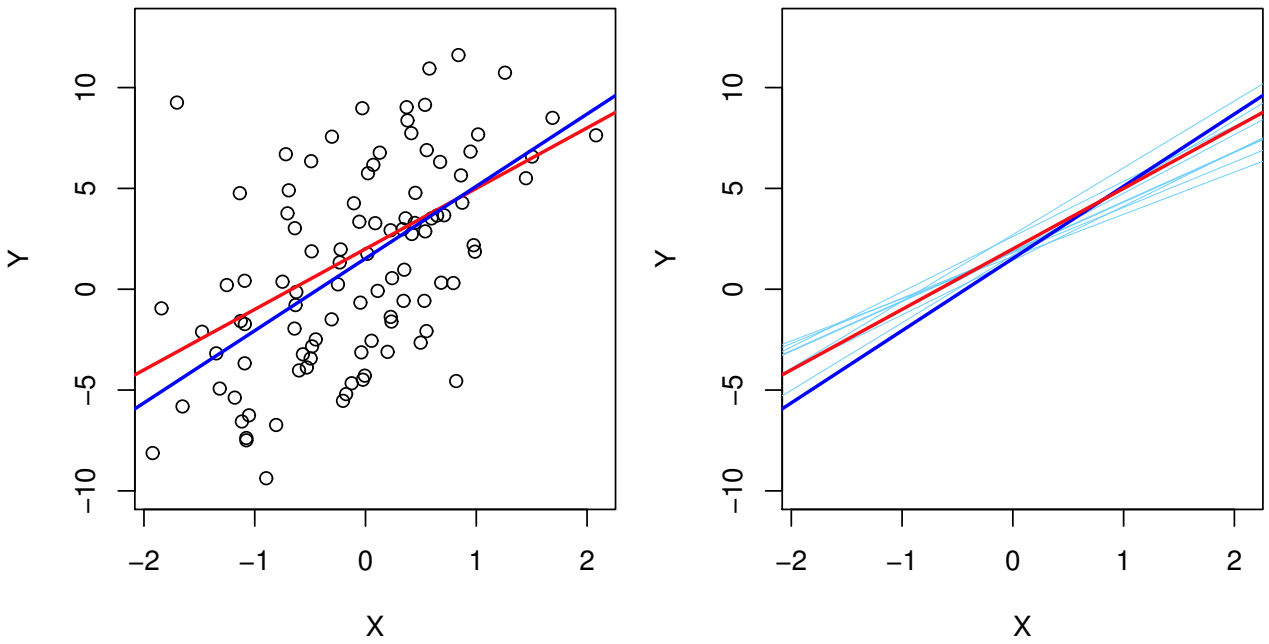
The estimated function: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

The residuals: $e_i = y_i - \hat{y}_i$

© Roberta Siciliano University of Naples Federico II

Population line versus Sample Fit

- *Modelling Identity*
  $y_i = E(Y|X = x_i) + \varepsilon_i$
- **POP line**
  $E(Y|X = x_i) = \beta_0 + \beta_1 x_i$
- *Fitting Identity*
  $y_i = \hat{y}_i + e_i$
- **SAMPLE fit**
  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

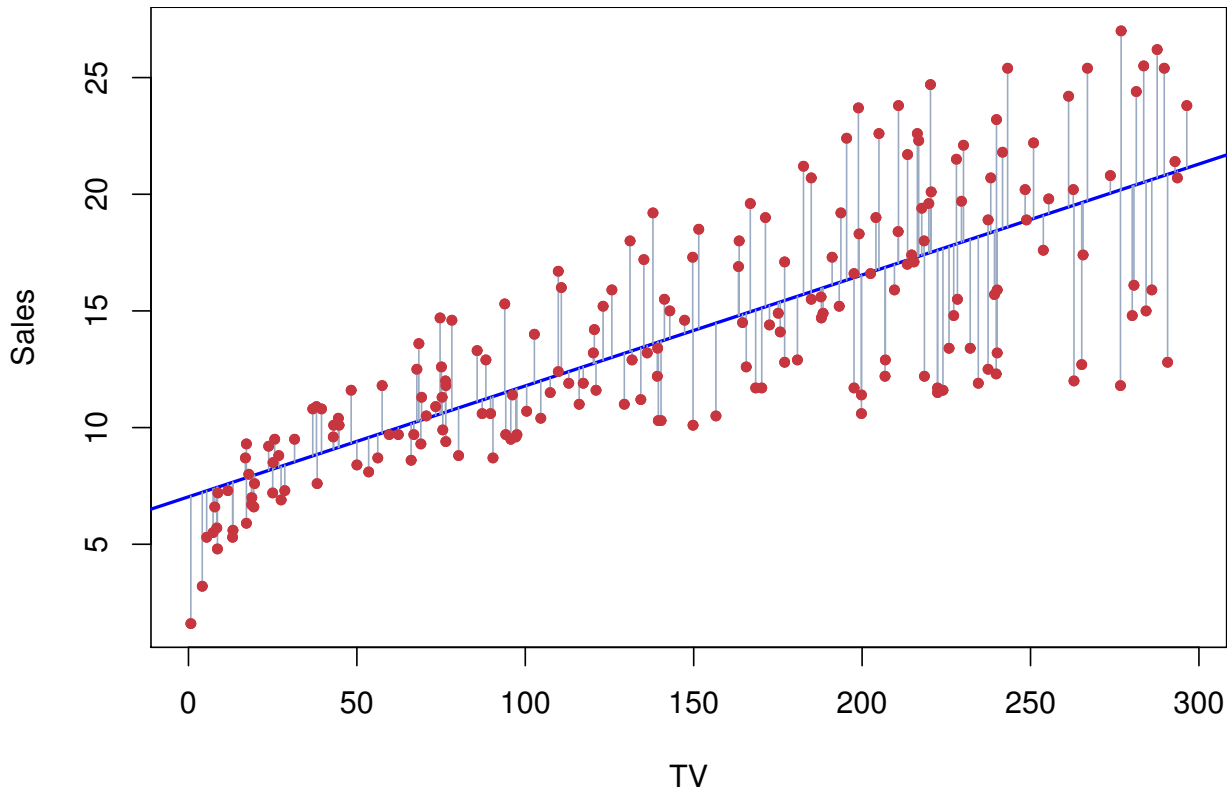


© Roberta Siciliano University of Naples Federico II

OLS fitting

Advertising data set:
Sales onto TV for 200 markets

$$\hat{y}_i = 7.03 + 0.0475x_i$$



© Roberta Siciliano University of Naples Federico II

The role of the residuals

Minimum of the loss function is the sum of squared residuals:

$$Q(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2 = \sum_{i=1}^{n} e_i^2 = RSS$$

where $0 \leq RSS \leq TSS$, specifically

- $RSS = 0$ when $y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i = \hat{y}_i$, for each i,
- $RSS = \sum_{i=1}^{n} (y_i - \bar{y})^2 = TSS$ (i.e., the total deviation of Y) when $\hat{y}_i = \bar{y}$, for each i, so that $\hat{\beta}_0 = \bar{y}$ and $\hat{\beta}_1 = 0$

The fitted model is so good as soon as the RSS is very low!

© Roberta Siciliano University of Naples Federico II

Goodness of fit

Start by the fitting identity, namely observed data is equal to the fitted model *plus* the error estimate (the residual):

$$y_i = \hat{y}_i + e_i$$

It is possible to derive:

$$y_i - \bar{y} = \hat{y}_i - \bar{y} + e_i$$

$$\sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n} e_i^2$$

Being $\sum_{i=1}^{n} (\hat{y}_i - \bar{y})\, e_i = 0$.

The **Total Deviation or Total Sum of Squares** $\boldsymbol{TSS}$ can be decomposed into **Explained Sum of Squares** $\boldsymbol{ESS}$ due to the regression model and **Residual Sum of Squares** $\boldsymbol{RSS}$:

$$TSS = ESS + RSS$$

Goodness of fit model:

$$R^2 = \frac{ESS}{TSS} = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS};\ 0 \leq R^2 \leq 1$$

© Roberta Siciliano University of Naples Federico II

Useful properties

- Sample covariance:

$$S_{XY} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- Sample Variances:

$$S_X^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad S_Y^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

- Cauchy-Schwarz inequality:

$$S_{XY}^2 \leq S_X^2 S_Y^2$$

- Sample linear correlation:

$$\hat{\rho}_{XY} = \frac{S_{XY}}{S_X S_Y} \qquad -1 \leq \hat{\rho}_{XY} \leq +1$$

- Regression coefficient estimate:

$$\hat{\beta}_1 = \frac{S_{XY}}{S_X^2} = \hat{\rho}_{XY}\frac{S_Y}{S_X}$$

✓ $R^2 = \hat{\rho}_{XY}^2 = \left(\hat{\beta}_1\right)^2 \frac{S_X^2}{S_Y^2}$

© Roberta Siciliano University of Naples Federico II

Accuracy of the estimates

OLS coefficient estimators are BLUE (Best Linear Unbiased Estimators by *the Gauss Markov Theorem*), normally distributed:

$$\hat{\beta}_0 \sim N\left(\beta_0, var\left(\hat{\beta}_0\right)\right), \hat{\beta}_1 \sim N\left(\beta_1, var\left(\hat{\beta}_1\right)\right)$$

with $var\left(\hat{\beta}_0\right) = \frac{\sigma^2}{n}\left(1 + \frac{\bar{x}^2}{s_X^2}\right)$ and $var\left(\hat{\beta}_1\right) = \frac{\sigma^2}{n}\frac{1}{s_X^2}$

The unbiased estimate of the variance of the errors σ^2 is given by

$$\hat{\sigma}^2 = \frac{\sum_i e_i^2}{n-2} = \frac{RSS}{n-2}$$

and $\hat{\sigma}$ is called Residual Standard Error (RSE).

Standard errors of the coefficient estimates:

$$SE\left(\hat{\beta}_0\right) = \frac{\hat{\sigma}}{\sqrt{n}}\sqrt{\left(1 + \frac{\bar{x}^2}{s_X^2}\right)} \quad SE\left(\hat{\beta}_1\right) = \frac{\hat{\sigma}}{\sqrt{n}}\frac{1}{s_X}$$

© Roberta Siciliano University of Naples Federico II

Testing procedures

- **Error Testing Variable:**

$$(n-2)\frac{\hat{\sigma}^2}{\sigma^2} \approx \chi^2_{n-2}$$

- **Model Testing Variable:** Under $H_0: \beta_1 = 0$

$$F = \frac{ESS/(2-1)}{RSS/(n-2)} = \frac{R^2}{(1-R^2)/(n-2)} \approx F_{(1),(n-2)}$$

- **Coefficient Testing Variables:**

Under $H_0: \beta_0 = b_0$ $\quad T_0 = \frac{\hat{\beta}_0 - b_0}{SE(\hat{\beta}_0)} \approx t_{(n-2)}$

Under $H_0: \beta_1 = b_1$ $\quad T_1 = \frac{\hat{\beta}_1 - b_1}{SE(\hat{\beta}_1)} \approx t_{(n-2)}$

For $H_0: \beta_1 = 0$: $(T_1)^2 = F$

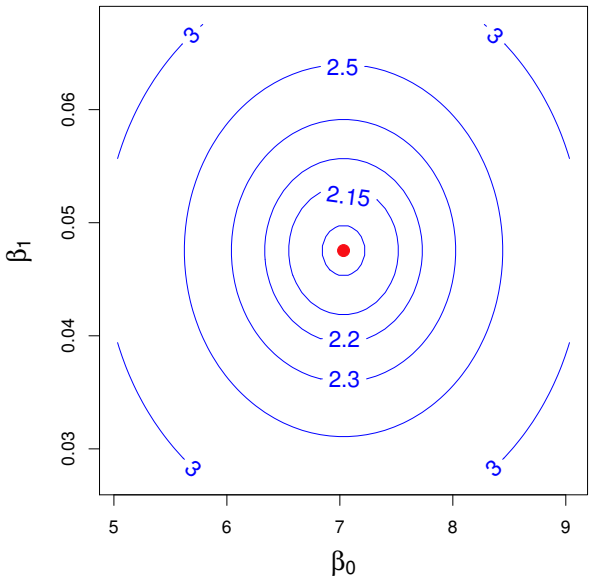
- **Confidence intervals of the regression coefficients:**

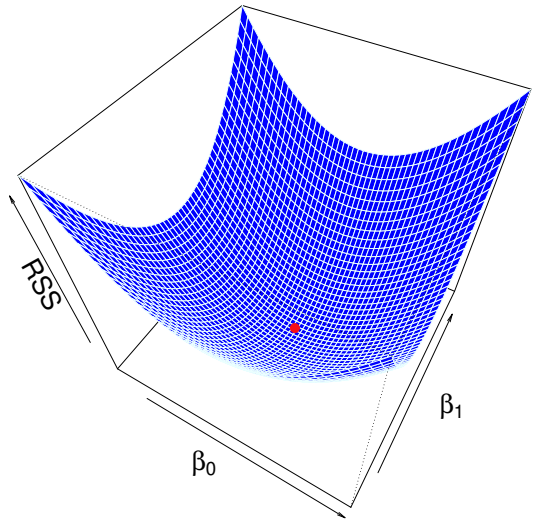
$$[\hat{\beta}_0 \pm t_{\alpha/2,(n-2)} SE(\hat{\beta}_0)] \; [\hat{\beta}_1 \pm t_{\alpha/2,(n-2)} SE(\hat{\beta}_1)]$$

© Roberta Siciliano University of Naples Federico II

Assessing the accuracy of the model

- **Advertising data set**: Sales onto TV for 200 markets
- Lack of fit: $RSE = 3.26$; $RSS = 2104.26$
- Goodness of fit: $R^2 = 0.612$
- Model testing: $F = 312.1$





		Coefficient	Std. Error	T-statistic	p-value
Under $H_0: \beta_0 = 0$	Intercept	7.0325	0.4578	15.36	<0.0001
Under $H_0: \beta_1 = 0$	TV	0.0475	0.0027	17.67	<0.0001

© Roberta Siciliano University of Naples Federico II

Running separate *simple* linear regressions

Advertising data set: Sales of a product (*expressed in (thousands of units*) in 200 different markets, along with advertising budgets (*expressed in thousands of dollars*) in each of those markets for three different media: TV, radio, and newspaper

	Coefficient	Std. Error	T-statistic	p-value
Intercept	7.0325	0.4578	15.36	<0.0001
TV	0.0475	0.0027	17.67	<0.0001

	Coefficient	Std. Error	T-statistic	p-value
Intercept	9.312	0.563	16.54	<0.0001
radio	0.203	0.020	9.92	<0.0001

	Coefficient	Std. Error	T-statistic	p-value
Intercept	12.351	0.621	19.88	<0.0001
newspaper	0.055	0.017	3.30	<0.0001

© Roberta Siciliano University of Naples Federico II

Prediction in *simple* linear regression

- How to estimate the value of Y for a new observation such that $X = x_0$?
- What about the prediction given by $\hat{\beta}_0 + \hat{\beta}_1 x_0$?
- $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ is *the prediction estimator of the systematic model component:*

$$Y_0 = f(x_0) = E(Y|X = x_0) = \beta_0 + \beta_1 x_0$$

  - Other notation can be also $\mu_{Y|X=x_0} = E(Y|X = x_0) = \beta_0 + \beta_1 x_0$.

- $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ is *the prediction estimator of the punctual value:*

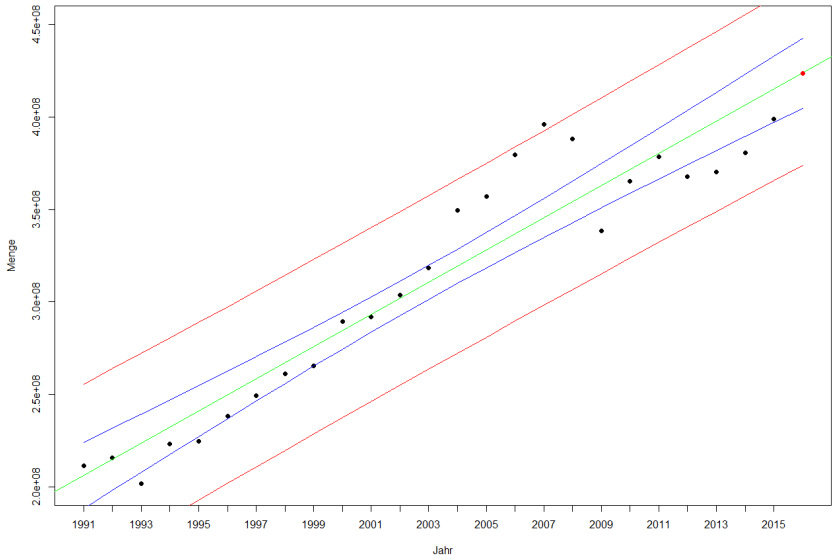
$$y_0 = f(x_0) + \varepsilon_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0$$

- Predictors $\hat{Y}_0$ and $\hat{y}_0$ are both BLUE and normally distributed estimators of Y_0 and y_0 respectively. They are unbiased $E(\hat{Y}_0) = E(\hat{y}_0) = \beta_0 + \beta_1 x_0$ but with a fundamental difference in their variance:

$$var(\hat{Y}_0) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_i x_i^2} \right) \qquad var(\hat{y}_0) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_i x_i^2} \right)$$

© Roberta Siciliano University of Naples Federico II

The standard errors of the estimators



Confidence Interval: $(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{\alpha/2,(n-2)}\hat{\sigma}\sqrt{\left(\frac{1}{n} + \frac{(x_0-x)^2}{\sum_i x_i^2}\right)}$

Prediction Interval: $(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{\alpha/2,(n-2)}\hat{\sigma}\sqrt{\left(1 + \frac{1}{n} + \frac{(x_0-x)^2}{\sum_i x_i^2}\right)}$

- The variance of the error σ^2 is estimated by $\hat{\sigma}^2 = RSS/(n-2)$.
- The *standard error of the predictor of the systematic component and of the punctual value* is respectively given by

$$SE(\hat{Y}_0) = \hat{\sigma}\sqrt{\left(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_i x_i^2}\right)}$$

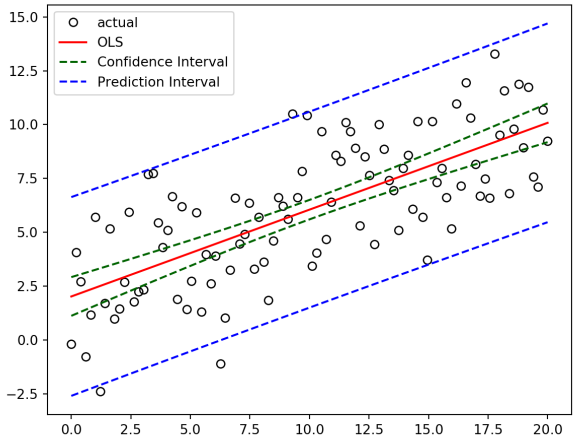
$$SE(\hat{y}_0) = \hat{\sigma}\sqrt{\left(1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_i x_i^2}\right)}$$

- where $SE(\hat{Y}_0) < SE(\hat{y}_0)$.
- The accuracy will be the greater the more it is close to the midpoint of coordinates $(\bar{x}, \bar{y})$, while the bands widen when we move away from this value so that the prediction error increases.
- The $(1-\alpha)\%$ ***confidence interval band*** for predicting Y_0 is much narrow with respect to the ***prediction interval band*** for predicting y_0 *which takes into account also of the variance of the error.*

© Roberta Siciliano University of Naples Federico II

Special cases

- Recall: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.
- At $x_0 = \bar{x}$ it holds that
  - $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$ with $var(\hat{Y}_0) = \frac{\sigma^2}{n}$.
  - $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} + \varepsilon_0$ with $var(\hat{y}_0) = \frac{\sigma^2}{n} + \sigma^2$.
- If there is no predictor, it holds that
  - $\hat{Y}_0 = \hat{\beta}_0 = \bar{y}$ and $var(\hat{Y}_0) = \frac{\sigma^2}{n}$
  - $\hat{y}_0 = \hat{\beta}_0 + \varepsilon_0$ with $var(\hat{y}_0) = \frac{\sigma^2}{n} + \sigma^2$.



© Roberta Siciliano University of Naples Federico II

Potential problems

Model mis-specification and Non-linearity

Correlation of error terms

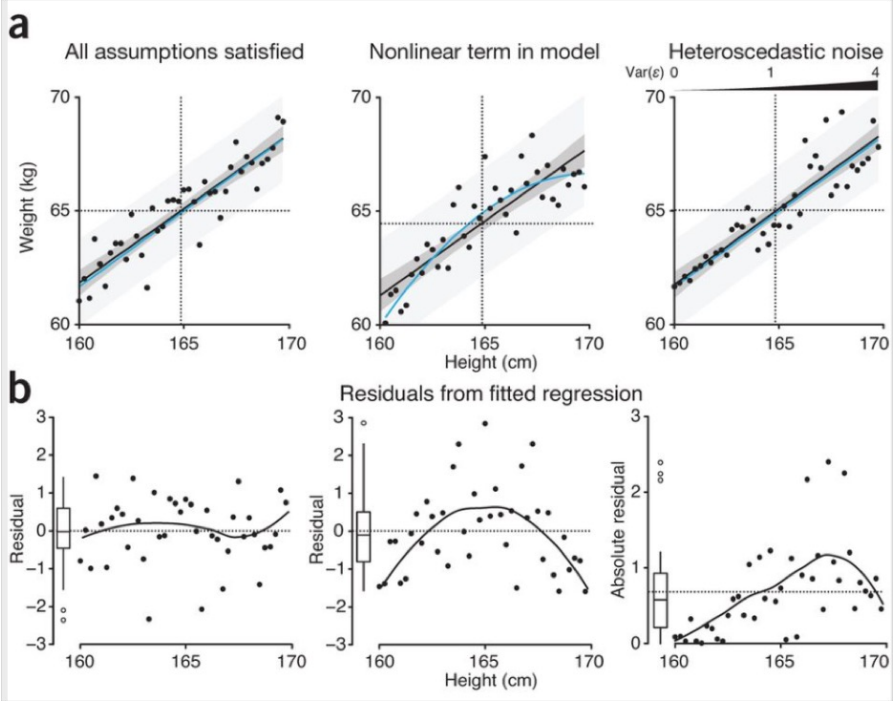
Non-constant variance of error terms

Outliers

High-leverage points

© Roberta Siciliano University of Naples Federico II

Regression diagnostics



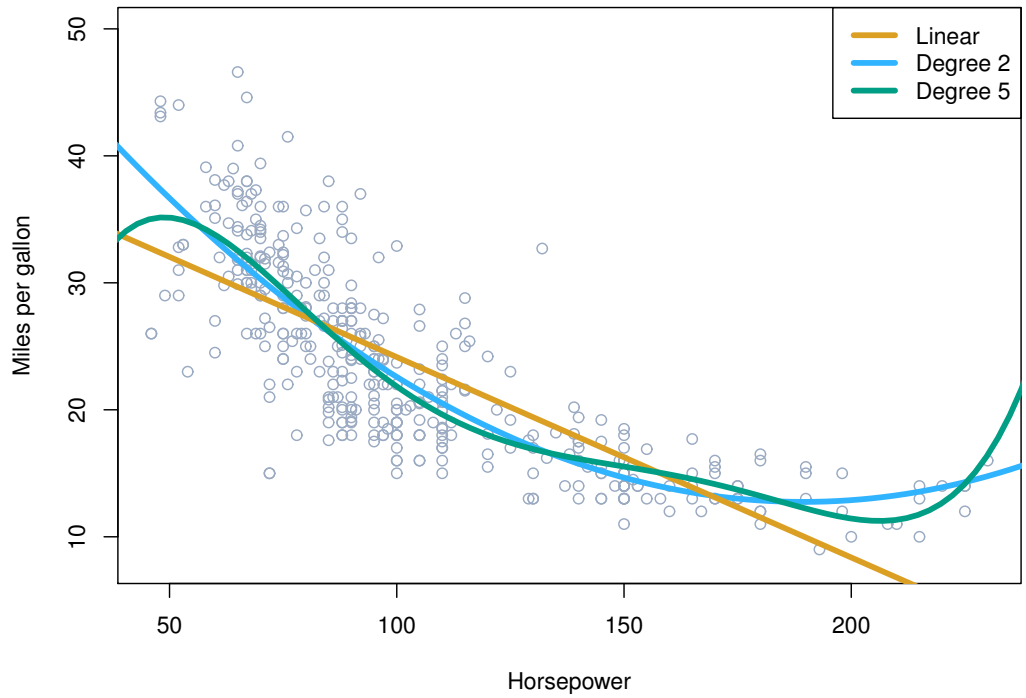
© Roberta Siciliano University of Naples Federico II

Non-Linear Relationships

Auto data set of a number of cars

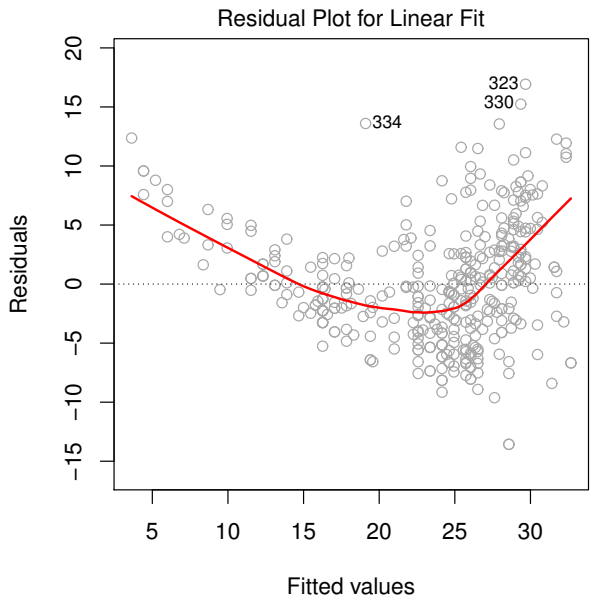
Mpg (gas mileage in miles per gallon) *versus*

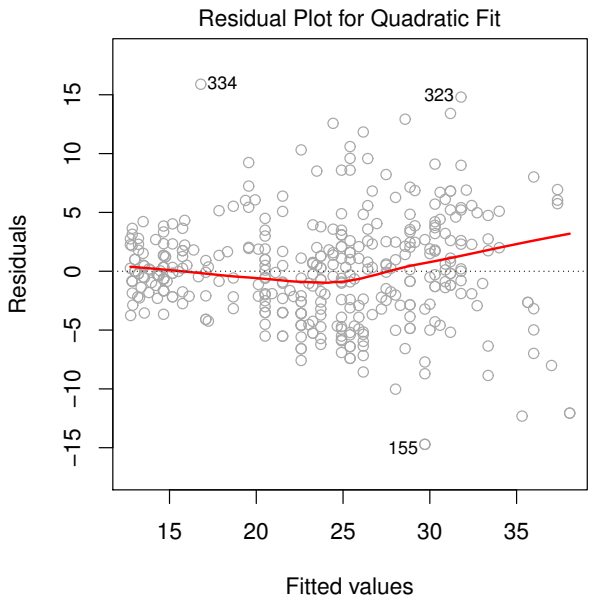
- ✓ **Horsepower**
- ✓ **Polynomial of Horsepower up to second-degree**
- ✓ **All polynomials of horsepower up to fifth-degree**



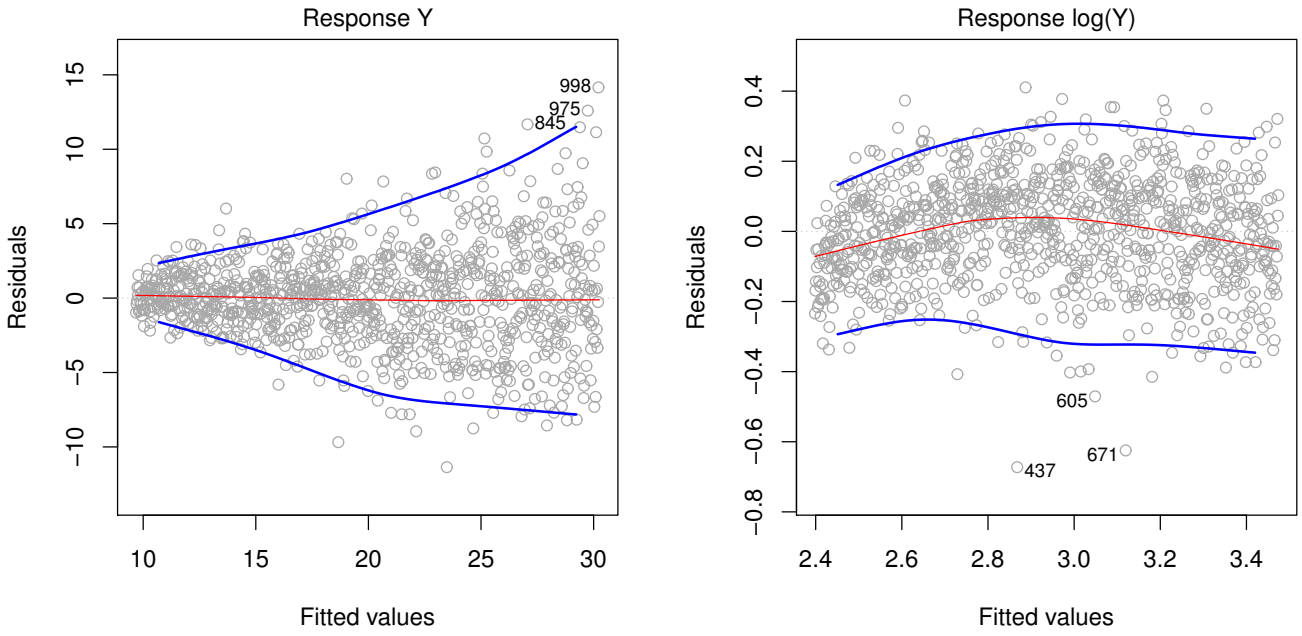
© Roberta Siciliano University of Naples Federico II

Residual Plots





© Roberta Siciliano University of Naples Federico II



Residual Plots to detect Heteroscedasticity

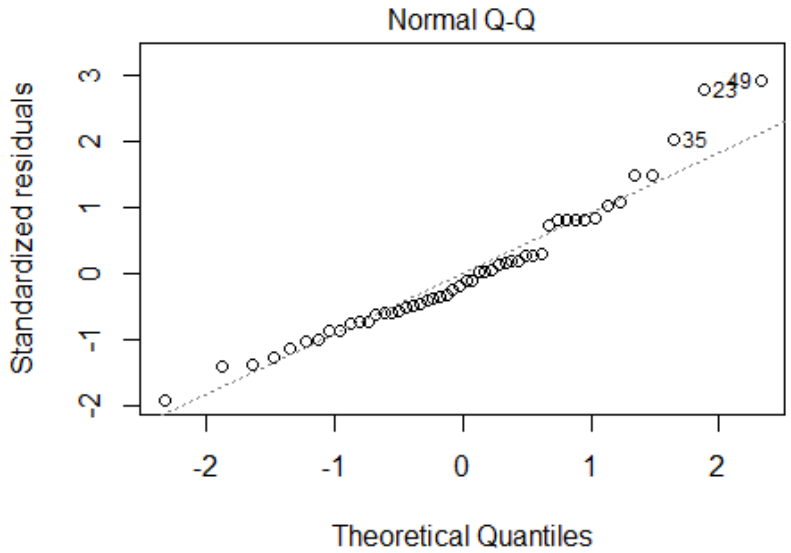
© Roberta Siciliano University of Naples Federico II

The Normal Probability Plot (QQ Plot)

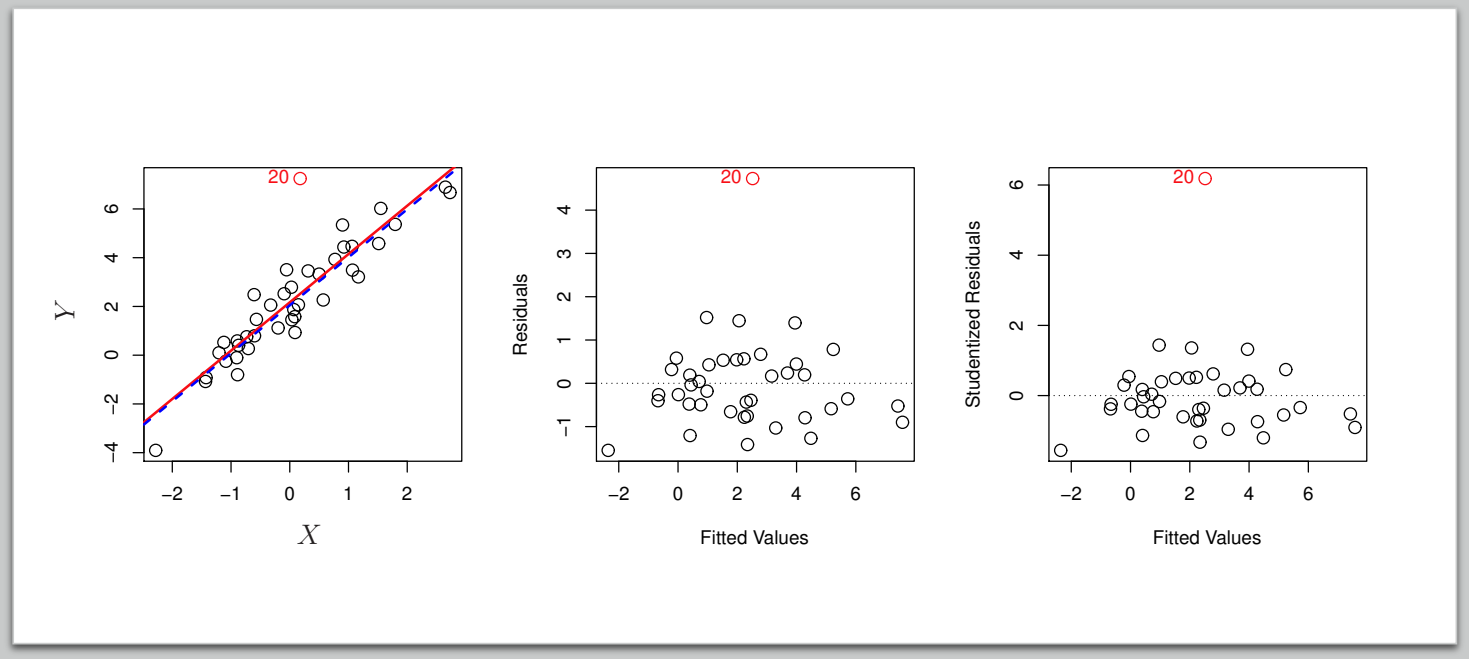
The best diagnostic test for normally distributed errors is a ***normal probability plot* or Q-Q plot of the residuals:** a plot of the quantiles of residual distribution *versus* the quantiles of a normal distribution having the same mean and variance.

If the distribution is normal, the points on QQ plot should fall close to the diagonal line.

- A ***bow-shaped*** pattern of deviations from the diagonal indicates that the residuals have excessive *skewness* (i.e., they are not symmetrically distributed, with too many large errors in the same direction).
- An ***S-shaped*** pattern of deviations indicates that the residuals have excessive *kurtosis*--i.e., there are either two many or two few large errors in both directions.



© Roberta Siciliano University of Naples Federico II

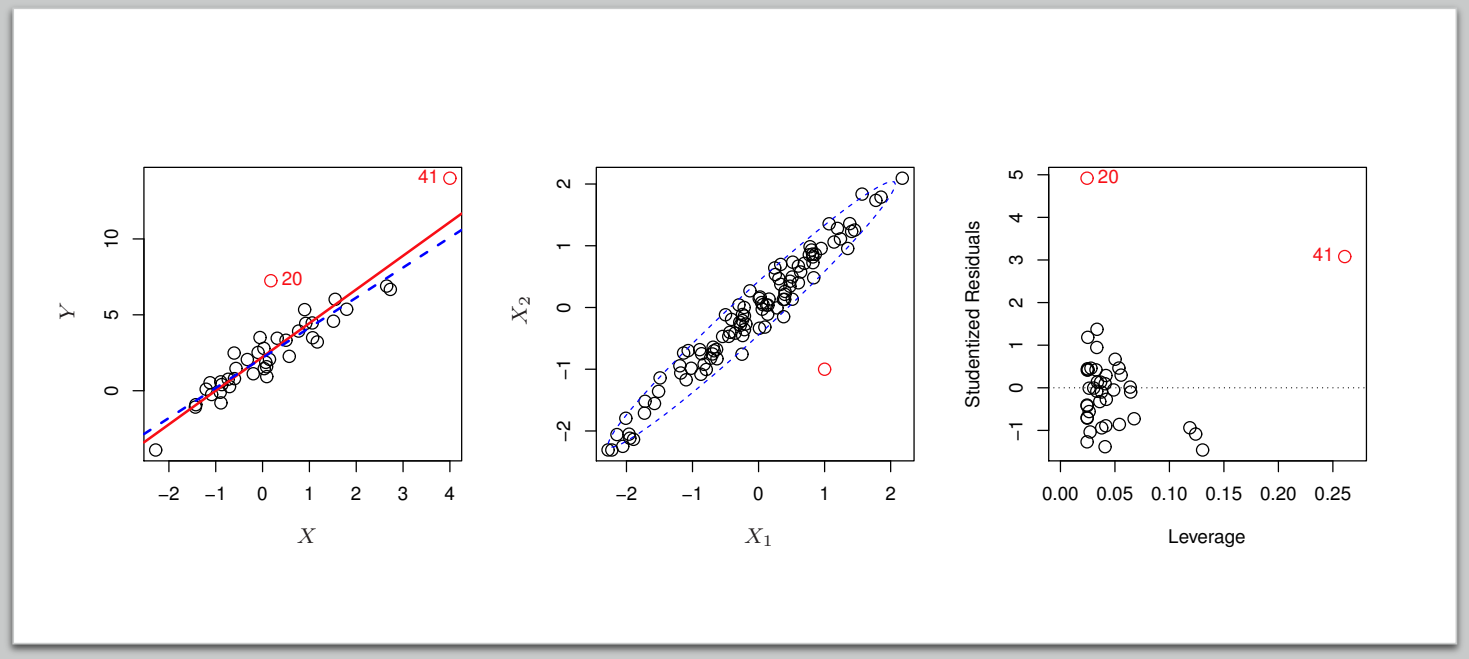


Outliers

An outlier is a point for which y_i is far from the value predicted by the model.

It causes reduction in the amount of explained variation to the fitted line as well as an increase in Residual Sum of Squares, thus reducing the R^2.

© Roberta Siciliano University of Naples Federico II



High-Leverage Points

Observations with high-leverage have an unusual value for x_i

The leverage statistic is $h_i = \frac{1}{n} + \frac{(x_i - x)^2}{\sum_{l=1}^{n}(x_l - x)^2}$

Cook's Distance:

$$D_i = \frac{\sum_{l=1}^{n}\left(\hat{y}_l - \hat{y}_{l(i)}\right)^2}{p \times \hat{\sigma}^2} = \frac{e_i^2}{p \times \hat{\sigma}^2} \times \left[\frac{h_{ii}}{(1 - h_{ii})^2}\right]$$

© Roberta Siciliano University of Naples Federico II